YU (CHARLIE) CHAN

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EDUCATION

Master of Science in Artificial Intelligence Jun 2022
National Yang Ming Chiao Tung University **GPA: [4.26/ 4.30]**
Exchange Semester at University of Agder, Norway Spring 2022
Bachelor of Science in Electrical Engineering and Computer Science Jun 2019
National Tsing Hua University

PROFESSIONAL EXPERIENCE

Research Assistant Mar 2026 - Present
Robot Learning Lab (RLLab), National Taiwan University — advisor: Prof. Shao-Hua Sun

- Lead the real-world experiments for “Implicit State Estimation via Video Replanning” (under review at CoRL 2026): built the physical evaluation stack — including designing and 3D-printing custom fixtures and task setups — and ran the real-robot experiments and ablations that extended an algorithmic contribution into physically validated results.
- Built the lab’s imitation-learning infrastructure integrating an AgileX Piper arm and a ROBOTIS OMY leader with the LeRobot stack — teleoperation, data collection, training, and evaluation — released three hardware plugins as open source, with Dynamixel motor support merged upstream into Hugging Face LeRobot (PR #3815).
- Researching tactile representation learning for manipulation: whether perception-level gains of tactile encoders transfer to policy-level success, using unpaired onboarding of new tactile sensors as controlled representation variation.

AI Application Engineer Nov 2024 - Mar 2026
Novatek

- Designed and optimized a dual-channel real-time sound event localization and detection (SELD) model and its inference pipeline for low-latency deployment on resource-constrained embedded systems.
- Built a real-time speech-to-text translation prototype targeting on-device deployment, covering streaming audio capture, recognition, and translation.
- Surveyed and prototyped VLM-based auto-labeling to assist and automate object-detection dataset annotation, reducing manual labeling effort for internal datasets.
- Evaluated parameter-efficient fine-tuning methods (e.g., LoRA) for adapting large models on edge devices, and delivered deployment recommendations adopted by the team.

Wi-Fi R&D Software Engineer Sep 2022 - Nov 2024
MediaTek

- Owned the Fine Timing Measurement (FTM) ranging feature end-to-end: rebuilt and optimized the firmware architecture, improving distance-measurement accuracy from 2–3 m to within 1 m on production Wi-Fi chipsets.
- Served as module owner of the RMAC firmware; designed and drove an architectural refactor that unified code across multiple product lines and eliminated config-based feature flags, cutting per-product maintenance and integration effort.
- Represented MediaTek at the Wi-Fi Alliance PlugFest (San Francisco) and an IoT interoperability event, debugging cross-vendor issues on site with customer and partner engineers.
- Achieved a 100% pass rate in ISTA certification tests, ensuring full compliance with Wi-Fi Alliance standards.
- Automated the end-to-end testing workflow with scripts covering regression and

certification runs, reducing manual testing time by 80%.

Co-Founder, Developer Jan 2021 - Present
Tera Thinker

- Led design and implementation of adaptive AI question-recommendation platform for high-school students, categorizing exam questions by topic and tailoring practices to cover individual knowledge gaps.
- Engineered a knowledge-tracing module to model student mastery over time, reinforcing weak areas and boosting engagement.
- Developed an end-to-end AI tutoring suite: integrated automated essay scoring for standardized Taiwanese tests, a guided-answering interface for scaffolded learning, and personalized exam-prep sessions with performance dashboards to track and optimize student progress; currently adopted by over 20 high schools.

PUBLICATIONS/CONFERENCE PRESENTATIONS

Chan, Y., Lin, P.-Y., Tseng, Y.-Y., Chen, J.-J., & Tseng, Y.-C. (2024). Learning-Based WiFi Fingerprint Inpainting via Generative Adversarial Networks. In *Proceedings of the 33rd International Conference on Computer Communications and Networks (ICCCN)*, Kailua-Kona, HI, USA, pp. 1–7. IEEE.

Ko, P.-C., **Chan, Y.**, Fu, Y.-H., Chen, C.-R., Chen, H.-A., Yuan, S.-X., Yeh, H.-J., Ma, W.-C., Du, Y., Mao, J., & Sun, S.-H. Implicit State Estimation via Video Replanning. Under review at *CoRL 2026*. arXiv:2510.17315.

RESEARCH EXPERIENCE

Master's Research Sep 2020 - Jun 2022
PAIRLab, National Yang Ming Chiao Tung University & XunweiTech (Huawei)

- Conducted M.S. thesis research on zero-shot Wi-Fi fingerprint inpainting under Professors Yu-Chee Tseng and Jen-Jee Chen in collaboration with XunweiTech
- Developed and validated a GAN-based inpainting model with a shared-weight discriminator, improving Wi-Fi fingerprint positioning accuracy by 20% over benchmarks.
- Extended research to Wi-Fi/MAG fingerprint augmentation using graph-based generative models, enhancing indoor positioning robustness and enabling real-world deployment.

Student Researcher Sep 2017 - Jun 2018
Behavioral Informatics & Interaction-Computation (BIIC) Lab, National Tsing Hua University

- Contributed to a doctor-child picture-book reading study for autism screening, focusing on textual modality in a multimodal (speech, text, image) framework.
- Developed a Python-based NLP pipeline using Jieba for Chinese word segmentation and keyword extraction to identify discriminative linguistic features.
- Designed and executed classification experiments, employing cross-validation to evaluate model generalizability and refine feature selection.

AWARDS

Book Scholarship/Presidential Award, National Yang Ming Chiao Tung University 2020
Top 25 Finalist, CHT 5G Application Competition 2021
Foxconn–NYCU Joint Research Center Scholarship 2021
vAward Recipient (×2), MediaTek 2023–2024

EXTRACURRICULAR SERVICE

Service-Learning Teaching Assistant 2017 - 2018
National Tsing Hua University Service-Learning

- Designed and delivered weekly English lessons for underprivileged students in Qionglin township for 2+ years, using engaging activities to boost proficiency and confidence.
- Mentored incoming volunteers, co-developed teaching materials, and led biweekly planning meetings to ensure curriculum quality.

Volunteer STEAM Tutor 2022

Community Church Tutoring Program

- Provided on-demand tutoring in math, science, and technology to high-school and university-bound students preparing for entrance exams.
- Crafted customized problem sets and study guides, improving students' problem-solving skills and exam readiness.

TECHNICAL SKILLS:

- Python, C, C++, PyTorch, Machine Learning, Deep Learning, GAN, LLM, ROS, Speech Recognition, AI in Education (advanced); Image Processing, TensorFlow, React, CAD/3D printing (intermediate)

LANGUAGES:

- Mandarin Chinese, Taiwanese (native); English (fluent)

RELATED COURSEWORK

Systems & Architecture

- Computer Architecture
- Advanced Computer Architecture
- Operating Systems

Algorithms & Theory

- Data Structures
- Design and Analysis of Algorithms
- Competitive Programming Training

AI & Robotics

- Introduction to Machine Learning
- Introduction to Image Processing
- Deep Generative Models
- ROS Robotics Programming and Projects